INCH POUND

MIL-H-7195G
7 October 1991
SUPERSEDING
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07 Feb 1986

MILITARY SPECIFICATION

HARDWARE, PARACHUTE, GENERAL SPECIFICATION FOR

This specification is approved for use by the Department of the Air Force and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the general requirements for all types of parachute hardware (see 6.4.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

FEDERAL

QQ-P-416	Plating, Cadmium (Electrodeposited)
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-636	Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Technology & Industrial Services Division, SA-ALC/TIRDM, Kelly AFB, TX 78241-5609 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 1670

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MILITARY

MIL-P-116	Preservation, Methods of
MIL-H-6875	Heat Treatment of Steel, Process for
MIL-A-8625	Anodic Coatings, for Aluminum and Aluminum Alloys
MIL-C-87115	Coating, Immersion Zinc Flake/Chromate Dispersion

STANDARDS

MILITARY

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	Marking for Shipment and Storage
MIL-STD-147	Palletized Unit Loads
MIL-STD-889	Dissimilar Metals
MIL-STD-1949	Inspection, Magnetic Particle
MIL-STD-2073/1	DoD Materiel Procedures for Development and Application of Packaging Requirements

(Unless otherwise indicated, copies of federal and military specifications, standards,and handbooks are available from the Standardization Documents Order Desk, Bldg 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ANSI/ASME B46.1	Surface Texture

(Application for copies of ASME publications should be addressed to American Society of Mechanical Engineers, 345 East 47th Street, New York, N.Y. 10017)

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AMERICAN SOCIETY OF FOR TESTING AND MATERIALS (ASTM)

ASTM A370	Steel Products, Mechanical Testing
ASTM D3951	Commercial Packaging
ASTM E18	Metallic Materials Rockwell Hardness and Rockwell Superficial Hardness of

(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia PA 19103.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified in the contract or purchase order, samples shall be subjected to first article inspection (see 4.3, 6.2.1 & 6.3).

3.2 Materials. Materials shall be as specified on the applicable detail drawing or military standard.

3.2.1 Metals.

3.2.1.1 Dissimilar metals. Unless suitably protected against electrolytic corrosion dissimilar metals shall not be used in contact with each. Dissimilar metals are defined in MIL-STD-889.

3.2.1.2 Heat treatment. When required, steel hardware shall be heat-treated in accordance with MIL-H-6875 or the applicable drawings or specifications. Unless otherwise specified on the drawing or standard, the applicable hardware which is made of alloy steel shall be heat treated to 170,000 psi to 190,000 psi tensile strength per MIL-H-6875.

3.2.1.3 Hardness. The Rockwell Hardness (RCH) values resulting from tests conducted in accordance with ASTM E18 shall be used as a measure of tensile strength of the applicable alloy steel hardware. The RCH range of values for the hardware shall be equivalent to the tensile strength range per ASTM A370 and allowances per ASTM E18. Unless otherwise specified on the drawing or standard, the overall equivalent RCH range for the tensile strength specified herein (see 3.2.1.2) shall be 36-43 RCH, which include allowances per ASTM E18 (See 4.4.2.5).

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3.3 Construction. The hardware shall be constructed in accordance with the applicable detail drawing or military standard (see 6.2). When this specification and the drawing/standard conflict, the drawing/standard shall govern.

3.3.1 Method of manufacture. Unless otherwise specified, the method of manufacturing of the hardware shall be by forging, stamping, forming, machining, casting, or by any other methods which meet specified requirements.

3.3.2 Hardware. Hardware shall be manufactured of material traceable to the original manufacturing source to confirm definitely that it complies with all requirements specified herein, particularly related to required physical properties of metals. Hardware shall not be surplus property or repaired.

3.4 Performance.

3.4.1 Proof loading. When specified on the applicable drawing or military standard, the hardware shall be subjected to the proof load specified thereon. No breakage or deformation beyond the dimensional or functional limits shall occur when tested per 4.4.2.3 and examined per 4.4.2.4.

3.4.2 Magnetic particle inspection. When specified in the applicable military standard or detail drawing, the hardware shall be subjected to magnetic particle inspection. Unless otherwise specified, no relevant indication of defects shall be present (see 4.4.2.7).

3.4.3 Functional. When specified on the applicable detail drawing or standard, the hardware shall pass all functional test and assembly requirements specified (see 6.4.2 & 4.4.2.4.2).

3.5 Dimensions. Hardware shall be in accordance with applicable drawings or military standards.

3.6 Surface roughness. Machined surfaces shall be as specified on the applicable detail drawing or military standard. Forged surfaces shall have a 115 roughness height rating or better except that surfaces adjacent to a flash line shall have a 135 roughness height rating or better.

3.7 Finish. The finish shall be as specified on the applicable drawings. All hardware shall be clean and free of scale, rust, discoloration, and deposits of any other foreign matter. Unless otherwise specified on the drawings, steel hardware parts (other than corrosion resistant steel) shall be cadmium plated per QQ-P-416, class 1, type I or zinc flake/chromate dispersion coated per MIL-C-87115, class 3. Aluminum hardware shall be anodized in accordance with MIL-A-8625, Type II, unless otherwise specified.

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3.8 Identification of product. Unless otherwise specified, all hardware items except those that are impractical to mark such as hook eyes, cables, and grommets shall be permanently marked with the manufacturer's symbol, the part number, and the year and month of manufacturer. In addition, hardware covered by detail drawings or military standards shall be marked with the number of the drawing or standard. If impression stamping is not used, the method of marking shall be subject to the approval of the procuring activity. The number shall be legible, separate, and distinctive from the manufacturer's code mark. Component parts of an assembly need not be marked unless the parts are functionally separable, such as separable links and release devices. All identification markings shall be in a location and applied in a manner that will not affect the serviceability of the part nor be detrimental to the corrosion protection of the hardware.

3.9 Workmanship. All details of workmanship shall be in accordance with the highest grade of manufacturing practice. The hardware shall be free from burrs, sharp edges, fins, pits, cracks, laps and burst. Inner forging flash shall be removed flush with adjoining surfaces. External forging flash or die shift irregularities shall be faired in with adjoining surfaces. All inner and outer edges shall be broken and smoothed. All surfaces shall be free from scale and not unduly roughened as a result of pickling. Except at the broken edges and rounded corners, surfaces fronting knurled faces of a sliding bar (i.e., quick-fit hardware) shall be flat.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of section 3 and 5. The inspection set forth in this specification shall become part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the

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contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When required (see 6.2), examination and testing of first article samples shall be made of completely fabricated items for all provisions of this specification.

4.4 Quality conformance inspection. Except as otherwise indicated, quality conformaance inspection sampling shall be performed in accordance with MIL-STD-105. Quality conformance inspection shall consist of the examinations and tests specified.

4.4.1 Inspection of materials and components. In accordance with 4.1, the contractor is responsible for insuring that components and materials used were manufactured, tested, and inspected in accordance with the requirements of referenced subsidiary specifications and standards to the extent specified herein or, if none, in accordance with this specification.

4.4.2 Inspection of the end item.

4.4.2.1 Visual examination. Hardware shall be visually examined to determine compliance with the applicable requirements of 3.7, 3.8 and 3.9.

4.4.2.2 Surface roughness. Surface roughness shall be in accordance with ANSI/ASME B46.1 as required by applicable drawing or military standard (see 3.6).

4.4.2.3 Proof loading. Each piece of hardware shall be subjected to proof loads specified on the applicable drawing or military standard (see 3.4.1).

4.4.2.4 Examination for dimensions and functional tests.

4.4.2.4.1 Dimensional. Hardware shall be in accordance with the dimensions specified on the applicable drawings or military standards or as specified by the procuring activity (see 3.5).

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4.4.2.4.2 Functional. The functional test shall be performed after proof loading. The hardware shall be examined for all functional test requirements specified on the applicable drawings or military standard or as provided by the procuring activity (see 3.4.3). The lot shall consist of units of hardware as defined by same part number. The sample size shall be the number of units indicated by inspection level S-4 except that no sample shall contain less than ten units. Failure of a unit to pass all functional test requirements shall cause rejection of the lot.

4.4.2.5 Hardness test. Sample units shall be tested to determine compliance within the range of hardness values (RCH) specified in 3.2.1.3. At least three valid indentations shall be made on each sample unit and the average value used in determining the harness number. The inspection level S-4 of MIL-STD-105 shall apply, except that each sample shall contain a minimum of ten pieces of hardware. Hardness testing shall be performed per ASTM E18 at location indicated on the applicable drawing or military standard or in the thickest sections suitable when not otherwise specified. Any piece in the sample having an equivalent tensile strength value that falls outside the specified tensile strength range shall cause rejection of the lot.

4.4.2.6 Packaging inspection. An examination will be made to determine if the preservation, packaging, packing and marking required in section 5 of this specification have been met. A sample unit shall be one shipping container fully prepared for delivery. The lot size shall be the number of containers in the inspection lot. The inspection level S-4 of MIL-STD-105 and the acceptable quality level of 4.0, shall apply. Defects shall be scored as follows:

Examination	Defect
Markings	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application.
Materials	Component missing; component damaged, or otherwise defective, affecting serviceability.
Workmanship	Inadequate application of components, such as incomplete closure of container flaps, loose strapping, inadequate taping or stapling. Bulging or distortion of containers.
Content	Number per interior package not in multiples of 10 each.
Weight	Weight of contents exceeds requirement.

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4.4.2.7 Magnetic particle inspection. When required (see 3.4.2), each piece of hardware shall be subjected to a magnetic particle inspection in accordance with MIL-STD-1949.

5. PACKAGING

5.1 Preservation and packaging. Preservation and packaging shall be level A or C or Commercial as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning. Parachute hardware shall be cleaned in accordance with MIL-P-116.

5.1.1.2 Drying. Parachute hardware shall be dried in accordance with MIL-P-116.

5.1.1.3 Preservative applications. Preservatives shall not be used.

5.1.1.4 Unit packs. Unless otherwise specified, each parachute hardware item shall be individually unit packed in accordance with method III of MIL-P-116.

5.1.2 Level C. The level C preservation for the parachute hardware shall conform to the MIL-STD-2073/1 requirements for this level.

5.1.3 Commercial. The commercial preservation of the parachute hardware shall be in accordance with the requirements of ASTM D3951.

5.2 Packing. Packing shall be level A, B or C, as specified (see 6.2).

5.2.1 Level A. The packaged parachute hardware items shall be packed in containers conforming to PPP-B-601, overseas type.

5.2.2 Level B. The packaged parachute hardware items shall be packed in containers conforming to PPP-B-636, class weather resistant.

5.2.3 Level C. The packaged parachute hardware items shall be packed in containers conforming to PPP-B-636, class domestic.

5.2.4 Palletized loads. When specified (see 6.2.1), palletized loads shall be effected in accordance with MIL-STD-147. Palletized loads shall be uniform in size and quantities to the greatest extent possible. If the containers are of a size which do not conform to any of the pallet patterns, use of each container must be approved by the contracting officer.

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5.2.5 Commercial. The packaged parachute hardware items shall be packed in containers conforming to ASTM D3951.

5.3 Marking for shipment. In addition to any special marking required by the contract or order, interior packages and exterior shipping containers shall be marked in accordance with MIL-STD-129.

5.3.1 Levels A, B and C. In addition to any special or other identification markings required by the contract each unit pack, intermediate, exterior, and unitized load shall be marked in accordance with MIL-STD-129.

5.3.2 Commercial. Commercial requirements shall be in accordance with ASTM D3951.

5.4 General.

5.4.1 Exterior containers. Exterior containers shall be of minimum tare and cube consistent with the protection required and shall contain equal quantities to the greatest extent practicable, of identical stock numbered items.

5.5 Packaging inspection. The inspection of these packaging requirements shall be in accordance with 4.4.2.5.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. Hardware covered by this specification is intended for use in all types of parachute systems.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.

b. Part number of the hardware required (see 3.3).

c. Whether first article sample is required (see 4.3).

d. When preservation, packaging and packing will be other than that specified in section 5.

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e. Selection of applicable levels of preservation and packaging and packing (see section 5).

f. Type of shipping container required for level A packing (see section 5).

6.3 First article. When required, the contracting officer shall include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results and disposition of first articles. Invitations for bids shall provide that the Government reserves the right to waive the requirement for first article testing to those bidders offering a product which has been previously approved or tested by the Government, and that bidders offering such products, who wish to rely on such production of test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract.

6.4 Definitions.

6.4.1 Parachute hardware. The term "parachute hardware" as used herein, shall be construed to be hardware suitable for all types of parachute systems. These parts are for securing or adjusting the harnesses and static lines, stiffening, locking, or shaping the pack and deployment bag, and/or for securing or mounting of components that make up the parachute assembly.

6.4.2 Functional use. Hardware consisting of moving parts is termed functional when all parts move freely and operate as intended on the drawing or standard. These would include such items as spring loaded guards on hooks passing over required diameter bar, locking and ejecting mechanisms, and slide bars for gripping webbing. When such operations of end items are obstructed in any way from performing the intended function within any specified loads or restricting dimension, these are determined not functional and therefore unacceptable end items.

6.5 Subject term (key word) listing.

Casting
Forging
Forging flash
Forming
Function test
Hardness
Aluminum

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Steel
Magnetic particle inspection
Plating, cadmium
Proof loading
Rockwell hardness
Stamping
Surface roughness

6.6 Changes from previous issue. Due to the extensive changes made to this specification, the margins of this specification are not marked to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content, and relationship to the last previous issue.

Custodian:
Air Force - 99
Navy - AS
Army - GL

Review activity:
Army - GL
Navy - AS
Air Force - 99

Preparing activity:
Air Force - 82

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-H-7195 G	2. DOCUMENT DATE (YYMMDD)

3. DOCUMENT TITLE

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME *(Last, First, Middle Initial)*	b. ORGANIZATION	
c. ADDRESS *(Include Zip Code)*	d. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON *(If applicable)*	7. DATE SUBMITTED *(YYMMDD)*

8. PREPARING ACTIVITY

a. NAME SA-ALC/TIRDM	b. TELEPHONE *(Include Area Code)* (1) Commercial (2) AUTOVON 945-6314
c. ADDRESS *(Include Zip Code)* CODE 82 KELLY AFB, TX 78241-5422	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340

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